

Lessons Learned Report

Project Name: Blockchain voting system **URL:**

<https://projectcatalyst.io/funds/11/cardano-use-cases-concept/blockchain-voting-system> **Project Number:** 1100039 **Project Leads:** David Tacuri

1. Blockchain does not replace existing systems, it complements them

One of the main lessons learned from the project is that blockchain technology should not be considered a replacement for traditional electoral systems, but rather as an additional layer of trust, audit, and integrity.

The developed system demonstrates that critical processes (scanning, digitization, quality control) must remain on traditional infrastructures for performance and operational reasons, while blockchain acts as an independent verification mechanism.

2. Balance between on-chain and off-chain is critical

It was identified that storing large volumes of information directly on the blockchain (such as votes per candidate or images) is neither efficient nor sustainable in terms of cost and performance.

Therefore, a hybrid approach was adopted:

- **On-chain:** hashes, states, references, and key metadata
- **Off-chain:** images (IPFS), detailed data, OCR processing

This model maintains security without compromising scalability.

3. Blockchain latency is a limiting factor

The use of Cardano showed that transaction confirmation times are not immediate, which limits its use in processes that require real-time responses.

This led to a design where:

- The operation continues outside the blockchain
- The blockchain records evidence afterwards

This pattern is fundamental for critical systems such as electoral processes.

4. Security does not depend only on blockchain

Although blockchain guarantees immutability, the project demonstrated that the overall security of the system depends on multiple layers:

- User access control
- Process state validation

- Integrity in data capture
- Security in APIs and backend

Blockchain is only one part of the total security model.

5. Traceability is one of the greatest contributions

The greatest value of the system is not the data registration itself, but the ability to fully reconstruct the process of a record:

- Who processed it
- At what moment
- What changes it underwent
- What evidence supports it

This significantly strengthens transparency and public trust.

6. Cost optimization is fundamental in blockchain

The analysis of the system volume (approximately 45,000 records with multiple candidates) identified that:

- It is not viable to register each individual piece of data on the blockchain
- It is necessary to group information or use hashes
- Design efficiency directly impacts costs and performance

This highlights the importance of designing with a scalability focus from the very beginning.

7. Hybrid architecture is the most viable approach

The project validated that the best approach is a distributed architecture that combines:

- Traditional backend (C#, APIs)
- Blockchain (Cardano)
- Decentralized storage (IPFS)

Each component fulfills a specific role within the system, achieving an optimal balance between performance, security, and decentralization.

8. Proof of Concept is key to validating hypotheses

The development of the PoC allowed moving from a conceptual idea to real validation, identifying:

- Technical limitations
- Necessary architectural adjustments
- Opportunities for improvement

This demonstrates that in blockchain projects, early experimentation is essential.

9. There is high potential for adoption in governments

The project confirms that technologies such as Cardano can be applied in governmental contexts, especially in:

- Electoral processes
- Identity systems
- Traceability of public information

However, its adoption also depends on regulatory, political, and organizational factors.

10. Future scalability requires new solutions (Hydra / Midnight)

It was identified that, to bring the system into large-scale production, it will be necessary to incorporate complementary technologies from the Cardano ecosystem such as:

- **Hydra:** to improve scalability and reduce latency
- **Midnight:** to incorporate privacy into sensitive data

These technologies represent the next evolutionary step of the project.

General Conclusion

The Blockchain Voting System project demonstrates that blockchain technology is a powerful tool for improving transparency and trust in electoral processes, provided it is used strategically within a hybrid architecture.

The lessons learned establish a solid foundation for future implementations at larger scale and in real government environments